



SAKSHAM GUPTA

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Location: 1-C, CK-5 Chanakya Rail Enclave, Chanakyapuri, New Delhi

EDUCATION

Manipal Institute of Technology, Manipal

Manipal, Karnataka • May 2027

B.Tech in Information Technology

CGPA: 9.16/10.0

Delhi Public School R.K. Puram

New Delhi • 2023

Class 12 – CBSE (90.2%)

Class 10 – CBSE (93.5%)

EXPERIENCE

Software Development Intern

June – July, 2025

Indian Railway Catering and Tourism Corporation (IRCTC)

- Analyzed high-throughput booking system (2M+ daily transactions), identifying latency bottlenecks in data flow and query execution
- Worked closely with relational databases to analyze query execution paths and optimize database query performance by 25%
- Documented database schemas, booking workflows, and data dependencies for system optimization initiatives
- Evaluated scalable system design patterns including microservices architecture for high-traffic applications
- Assisted in automating database performance and load tests using Python scripts to simulate concurrent transactions

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C++, C

Data Science & Machine Learning: Pandas, NumPy, Scikit-learn, EDA & Visualization (Matplotlib, Seaborn), Data Analysis, Model Evaluation & Validation

Databases, Tools & Security: SQL, MongoDB, Git, VS Code, IntelliJ IDEA, Docker

Web Technologies: Flask, Node.js, React, REST APIs, HTML/CSS

Software Engineering: SDLC, Agile Methodology, Requirements Engineering, Software Testing, UML Diagrams, Risk Management

Systems & Performance: Performance Optimization, System Design, Profiling & Debugging

PROJECTS

SafePay | Machine Learning Based Transaction Fraud Detection System | [GitHub](#) • [View Live](#)

- Built an ML-based fraud detection model with data preprocessing and feature engineering, achieving 85%+ fraud detection accuracy
- Integrated real-time model inference into a Python backend (Flask) with MongoDB for transaction validation
- Implemented end-to-end security using AES-256 encryption, digital signatures, and secure password hashing

Fileshare | Real-Time File & Text Sharing | [GitHub](#) • [View Live](#)

- Built a full-stack real-time application using React frontend and Node.js/Express backend with Socket.io for live communication
- Implemented REST APIs and room-based session logic with automatic expiry for privacy and scalability
- Deployed production-ready system to cloud with CI/CD integration and structured error handling

Starfinder | Multi-Class Stellar Classification System | [GitHub](#) • [View Live](#)

- Developed multi-class stellar classification using Random Forest, achieving 93% accuracy across 6 star types
- Implemented domain-aware validation pipeline to detect out-of-distribution inputs and prevent invalid compact object predictions
- Deployed production ML system with Flask REST API, automated model persistence, and comprehensive EDA pipeline

Code2Flow | AI Code-to-Flowchart Generator | [GitHub](#) • [View Live](#)

- Built an AI-powered tool that converts reactor code into deterministic ASCII-based flowcharts using Python (Flask) and the Gemini API
- Designed a hybrid pipeline separating LLM-based control-flow extraction from deterministic Python rendering for stable output
- Established keyword-based normalization and validation to handle unreliable LLM formatting and minimize API usage

LEADERSHIP & ACTIVITIES

Organizing Committee Member — Sports (Chess)

March 2025

Revels'25, MIT Manipal

- Led logistics coordination for chess tournaments with 200+ participants, managing registration and bracket systems
- Collaborated with a 12-member cross-functional team ensuring 100% on-time event execution during cultural fest

ADDITIONAL

Languages: Hindi (Native), English (Proficient)

Interests: Competitive Programming, Chess, Badminton, Reading

ACHIEVEMENTS

Achievers Scholarship (Top 5% in IT)

Active contributor to open-source projects